How to upgrade from Xamarin to .NET 7

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MODERN CLIENT

David Ortinau

Principal Product Manager

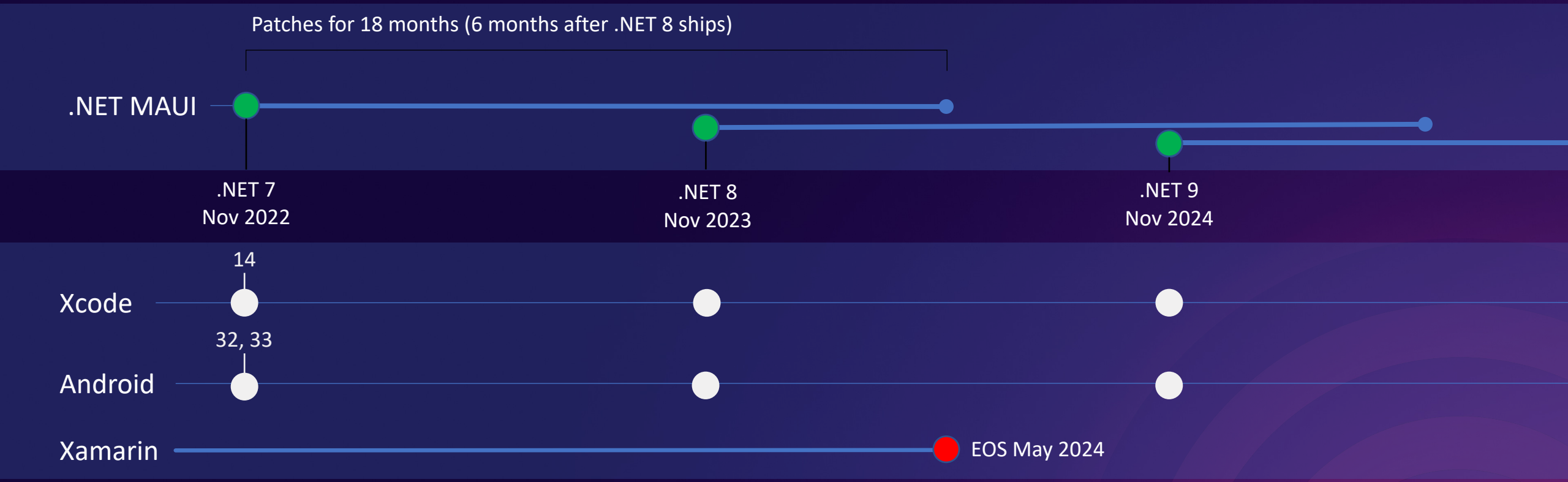
St. Louis, Missouri USA

Help customers succeed delivering client app experiences across various screens using .NET SDKs for mobile and desktop cross-platform – aka .NET MAUI (and Xamarin).

I like to trail run and have completed multiple distances up to 50 miles.

Grandfather. Doesn't age.

Release schedule, support & patching



- A major version of .NET MAUI will ship with .NET going forward (.NET 7, .NET 8, etc.)
- .NET MAUI is an optional workload with external dependencies (such as Xcode & Android SDK tools)
- A major version of .NET MAUI will receive patches for 6 months after a successor (major version) ships
- Xamarin SDKs will receive support until May 2024

Resources

- Discussions
  https://github.com/dotnet/maui/discussions/categories/upgrading-from-xamarin

- Documentation
  https://learn.microsoft.com/dotnet/maui/migration/

I have a Xamarin app so...

Why should I upgrade?

Benefits of Upgrading

	Xamarin	.NET MAUI	Benefit
C# version	7.3	10+	Latest language features
.NET SDK/Runtime	.NET Framework 4.8	.NET 6+	Latest performance and features
Android, iOS, Windows	API 32, Xcode 14, UWP	API 32+, Xcode 14+, WinUI3+	Modern platform versions
Platform customization	Custom Renderers	Handler hooks	Renderers still work, but handler overrides are much easier to write and maintain

759ms

Android feature rich app startup (33% faster)

175%

Faster UI rendering

Additional Benefits

- Smaller apps on disk
- Single project
  - Easier NuGet management
  - One set of images, app icon, splash
  - One place for fonts and assets
- Easier customization of platforms
- Desktop support
- Blazor hybrid
  - More code share, skill share
  - 100s of UI components
- CLI support
- .NET hot reload (C#, XAML, Razor, CSS)

Mobile Developer Timeline

Final versions supported

- Xamarin.iOS with Xcode 14.x
- Xamarin.Android with API 33



Today

September 2023

Xcode 15 GA
- New apps must use 15
- Upgrades may use 14

April 2024

Xcode 15
- ALL apps must use 15

May 1, 2024

Xamarin End of Support

The Upgrade

From Xamarin to .NET

The Move

Xamarin	.NET
.NET Framework	.NET 7
Xamarin.Android	.NET for Android
Xamarin.iOS	.NET for iOS
Xamarin.Mac	.NET for macOS
Xamarin.tvOS	.NET for tvOS
Xamarin.Forms	.NET MAUI
Franken-project	SDK style project
Multiple projects	Multiple projects OR single project
Binding library project	SDK style project
using Xamarin.Forms.*	using Microsoft.Maui.*
monoandroid12	net7.0-android
Xamarinios20	net7.0-ios
etc.	etc.

Xamarin.Android

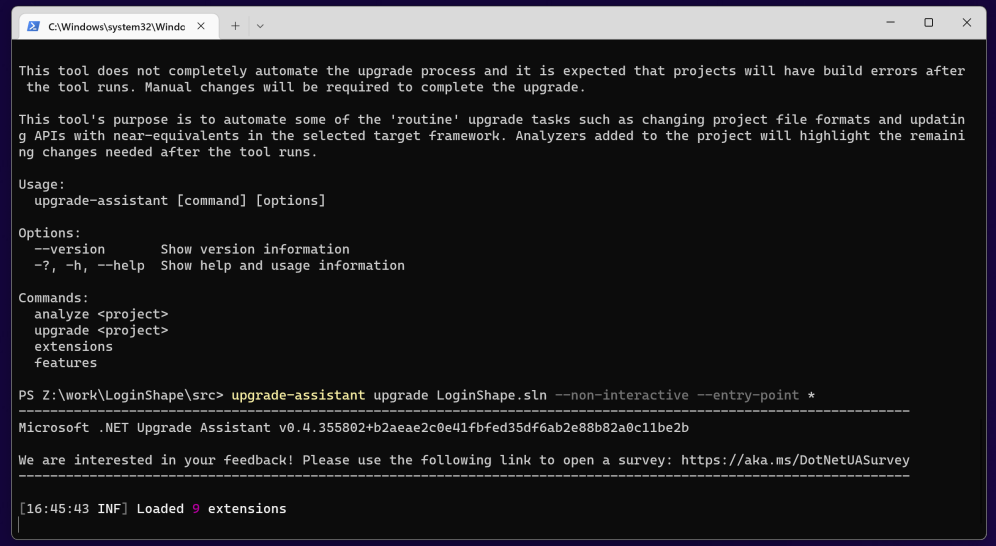
- Upgrade the project type and recompile
- Replace any incompatible dependencies
  - they might all just work!

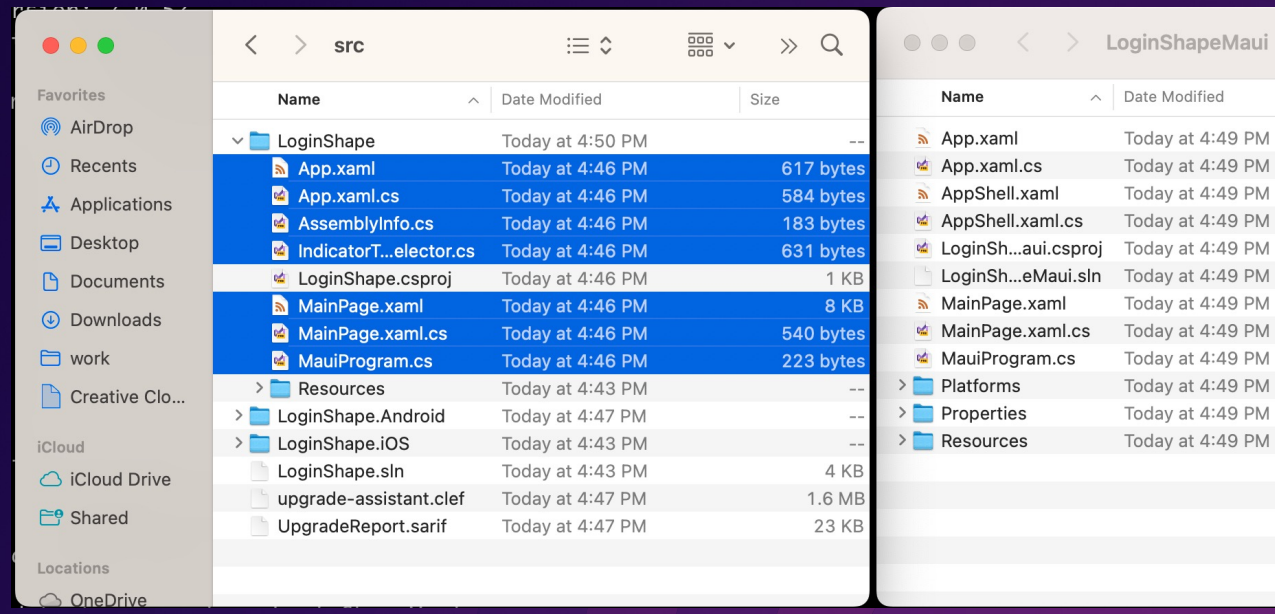
Xamarin.iOS

- Upgrade the project type
- Replace all dependencies with .NET 6+ versions
- Recompile and address errors

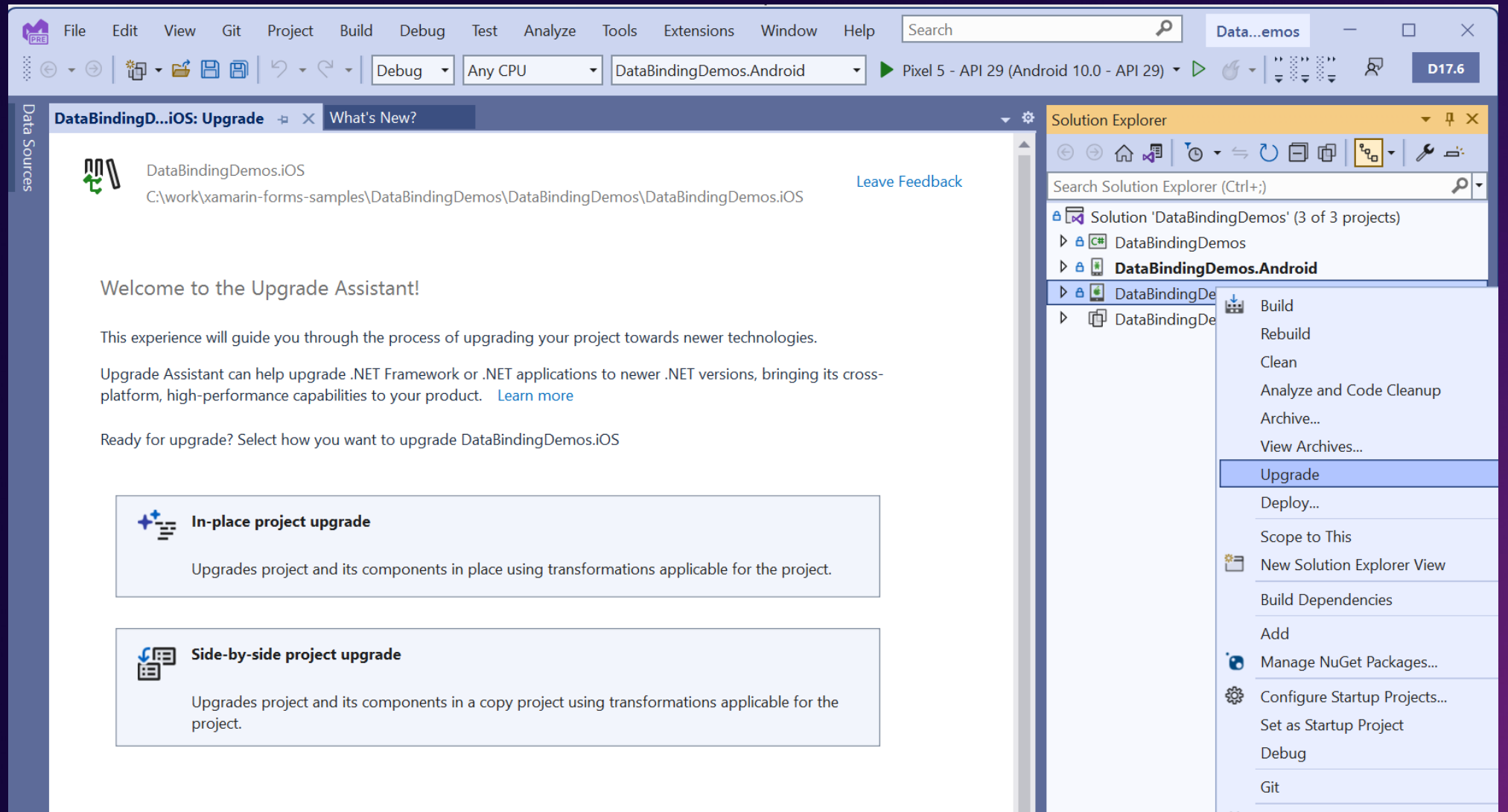
Two Options

- .NET Upgrade Assistant
- Copy Pasta



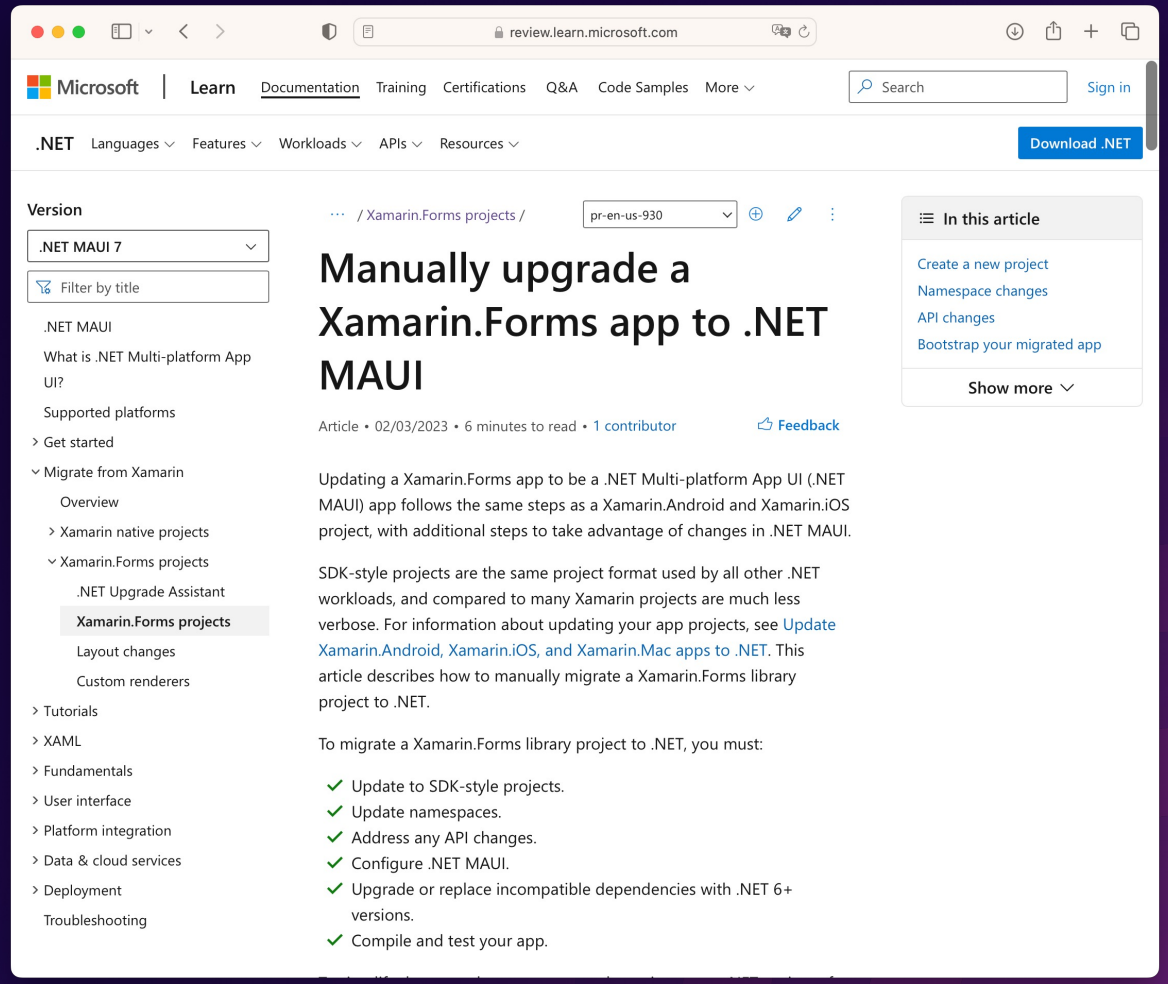


.NET Upgrade Assistant extension with Visual Studio 17.6

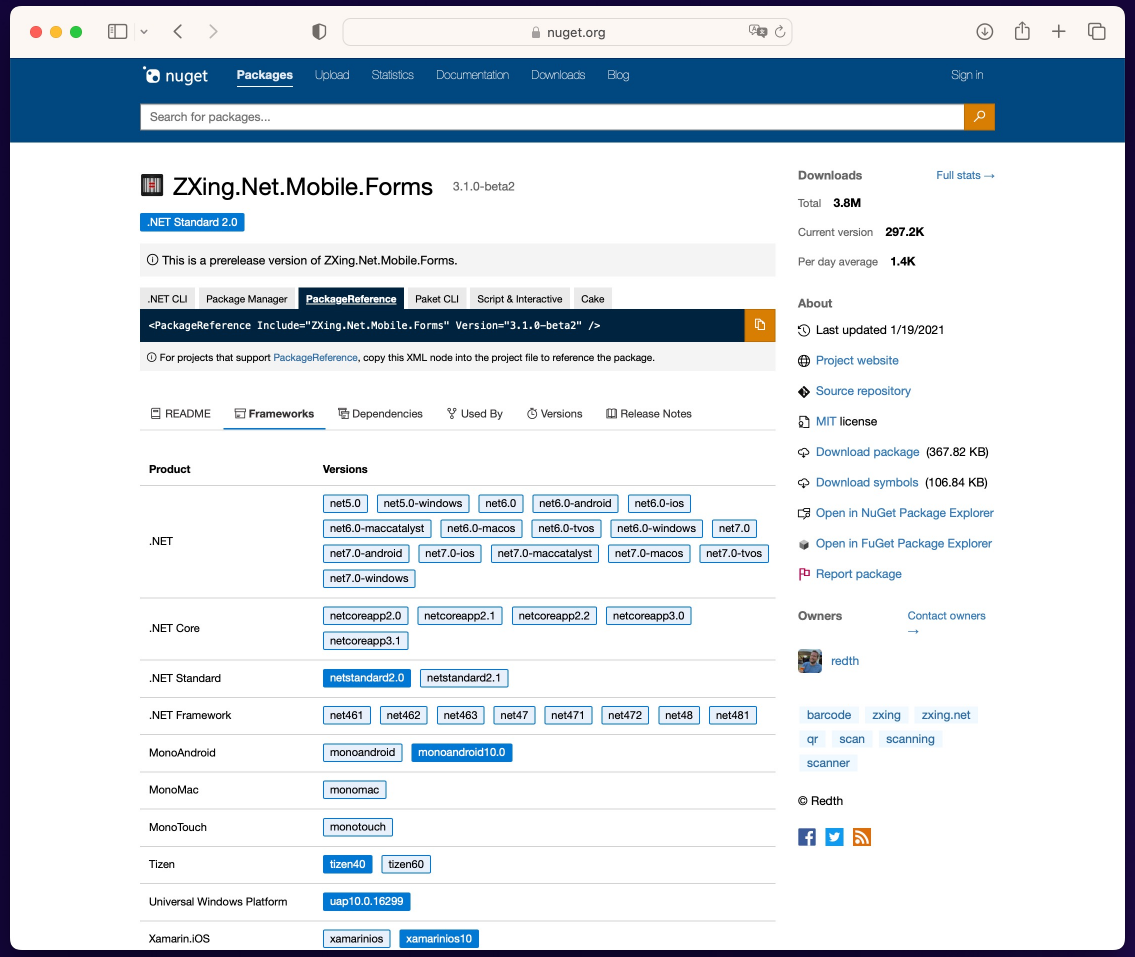


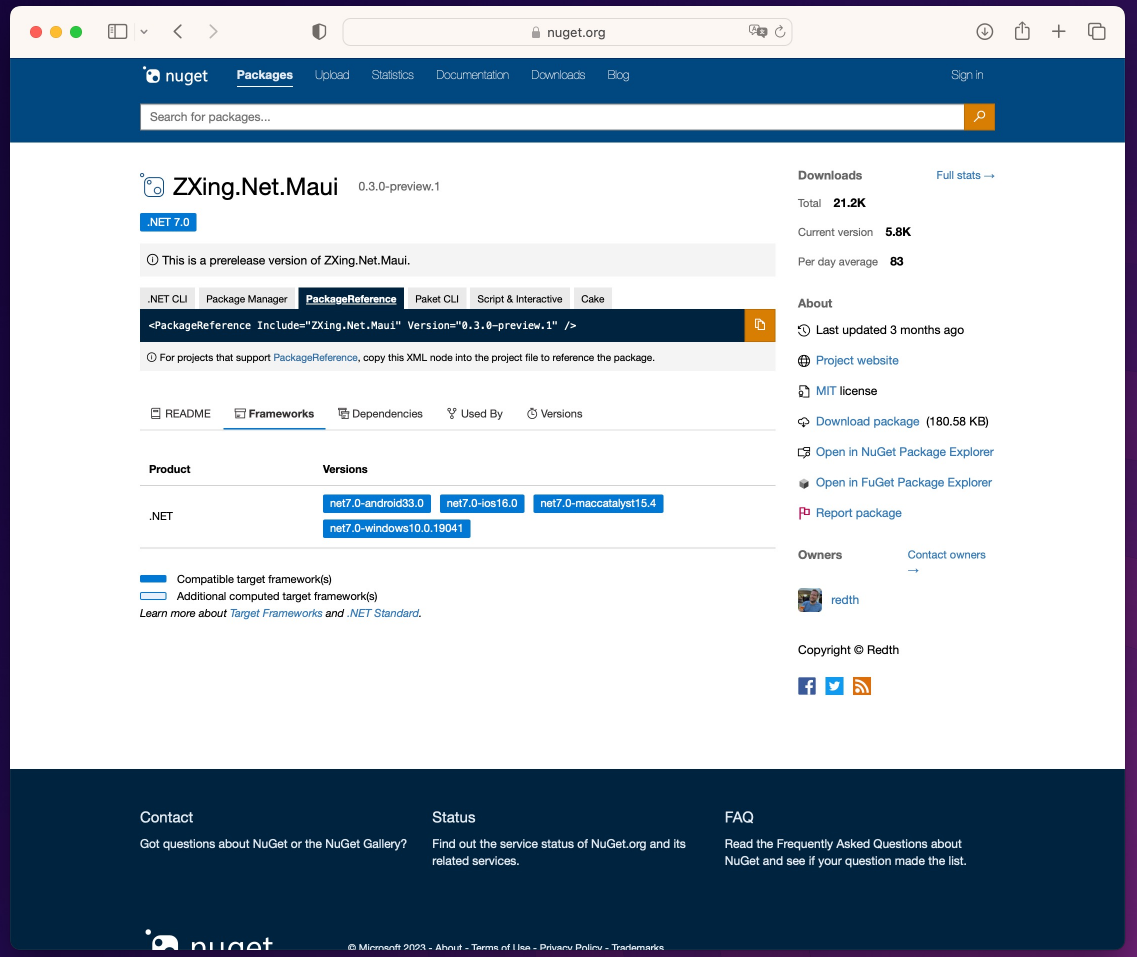
Upgrade tasks

- Update to SDK-style projects.
- Update namespaces.
- Address any API changes.
- Configure .NET MAUI.
- Upgrade or replace incompatible dependencies with .NET 6+ versions.
- Compile and test your app.

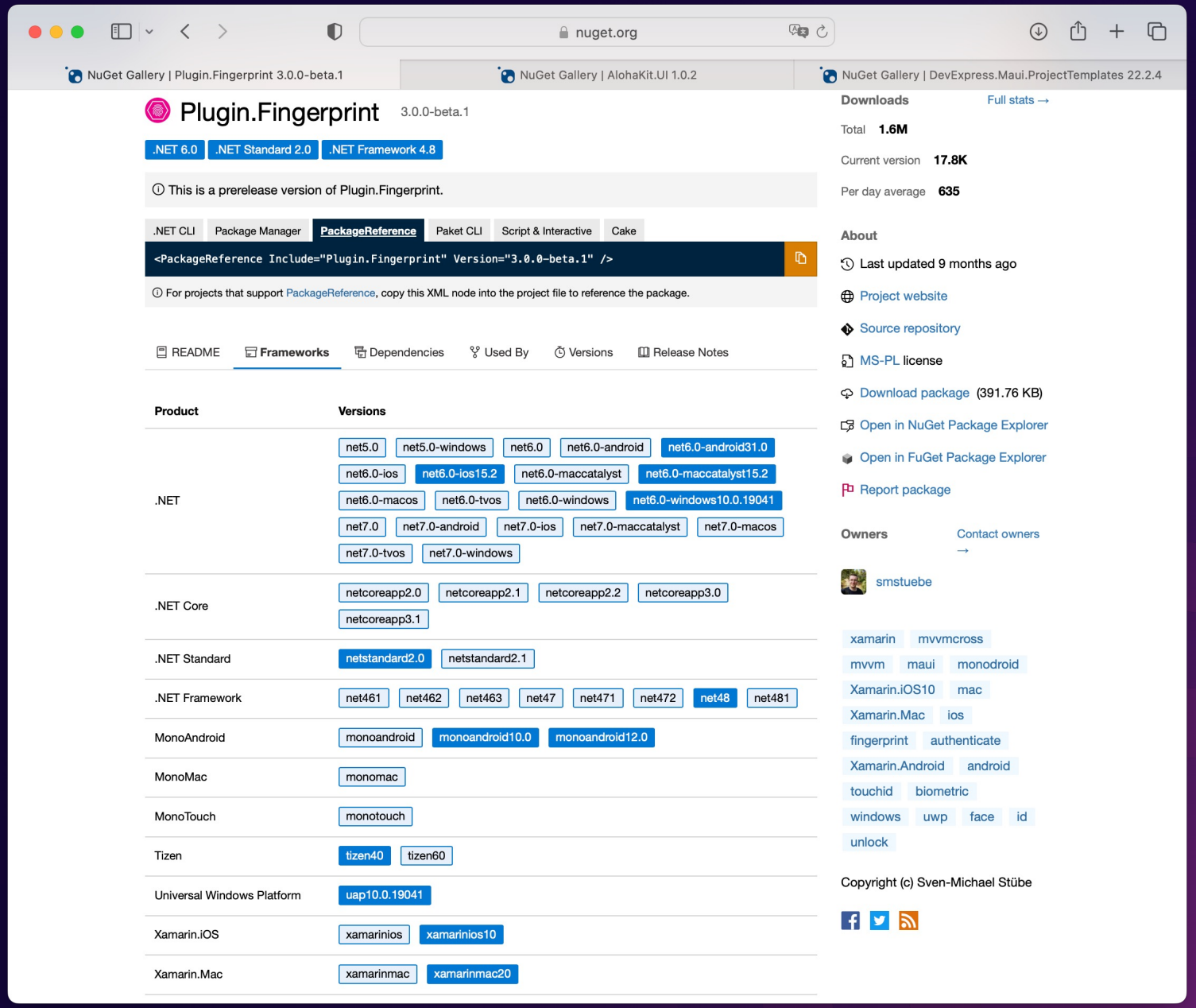


Which Nuget?

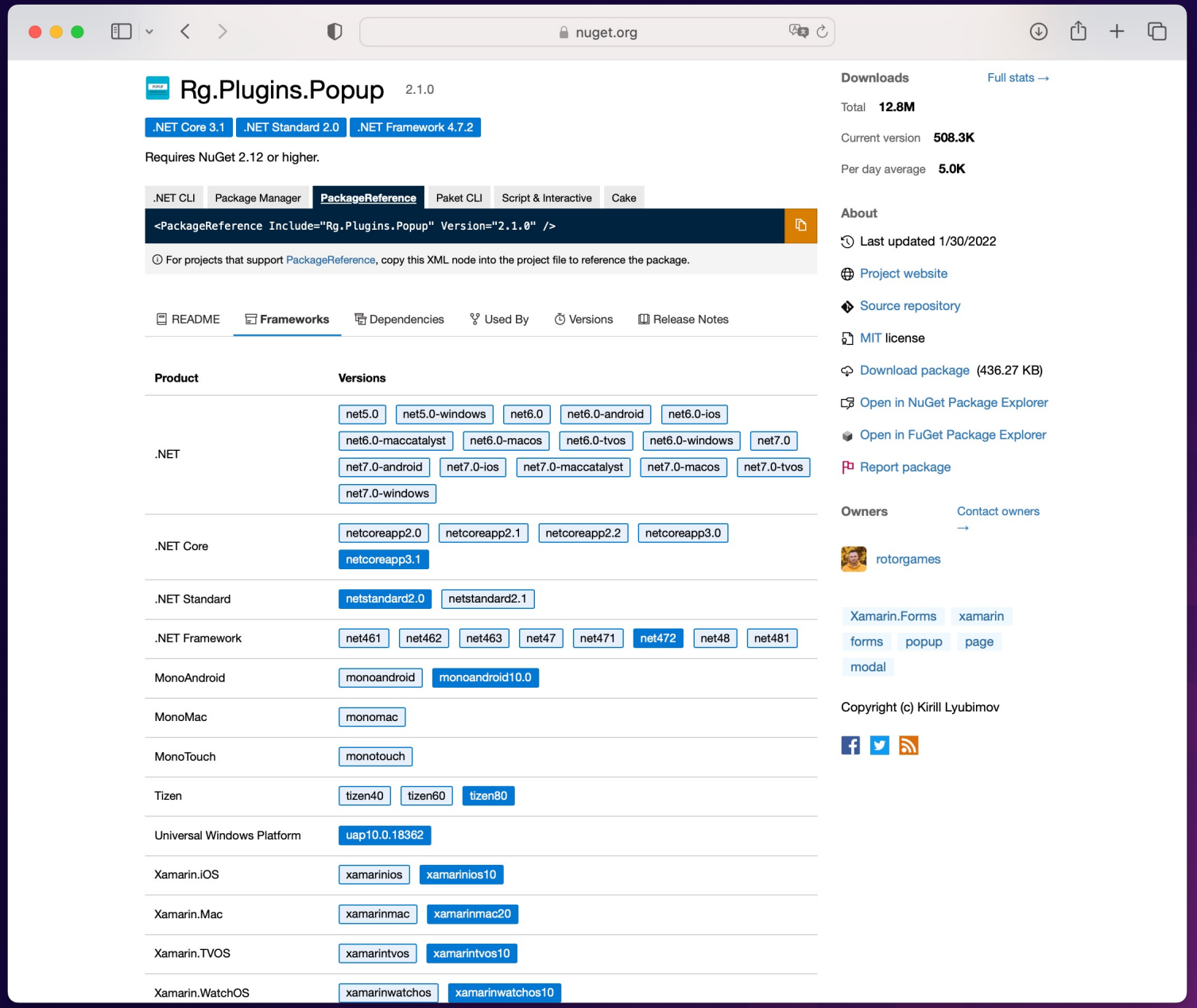

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.NET Standard
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NuGet Gallery | DevExpress.Maui.ProjectTemplates 22.2.4
Plugin.Fingerprint 3.0.0-beta.1
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.NET Standard 2.0
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Rg.Plugins.Popup 2.1.0
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.NET Standard 2.0
.NET Framework 4.7.2
Requires NuGet 2.12 or higher.
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<PackageReference Include="Rg.Plugins.Popup" Version="2.1.0" />
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Layouts

Behavior changes

Layout	Xamarin.Forms	.NET MAUI	Recommendation
All	In certain cases sizing requests would not be honored.	Sizing requests are honored.	
Grid	Columns and rows can be inferred from XAML.	Columns and rows must be explicitly declared.	Add ColumnDefinitions and RowDefinitions.
HorizontalStackLayout		"*AndExpand" has no effect.	
RelativeLayout		Requires the compatibility namespace.	Use Grid.
ScrollView	Arbitrary minimum size, inconsistent sizing	Expands to fill all space	Set a size if within a stack to enable scrolling.
StackLayout	Children can fill space in the stacking direction.	Children are stacked and will go beyond available space.	If you need child views to fill space, change to a Grid.
VerticalStackLayout		"*AndExpand" has no effect.	

Value changes

Property	Xamarin.Forms value	.NET MAUI value
Grid.ColumnSpacing	6	0
Grid.RowSpacing	6	0
StackLayout.Spacing	6	0

StackLayout

- Very simple – stacks in a single direction until all children stacked.
- Does NOT subdivide space
- Remove any "AndExpand" properties which are orthogonal to the stacking direction.

XAML Copy

```
<StackLayout>
    <Label Text="Hello world!"/>
    <Image VerticalOptions="FillAndExpand" Source="dotnetbot.png"/>
</StackLayout>
```

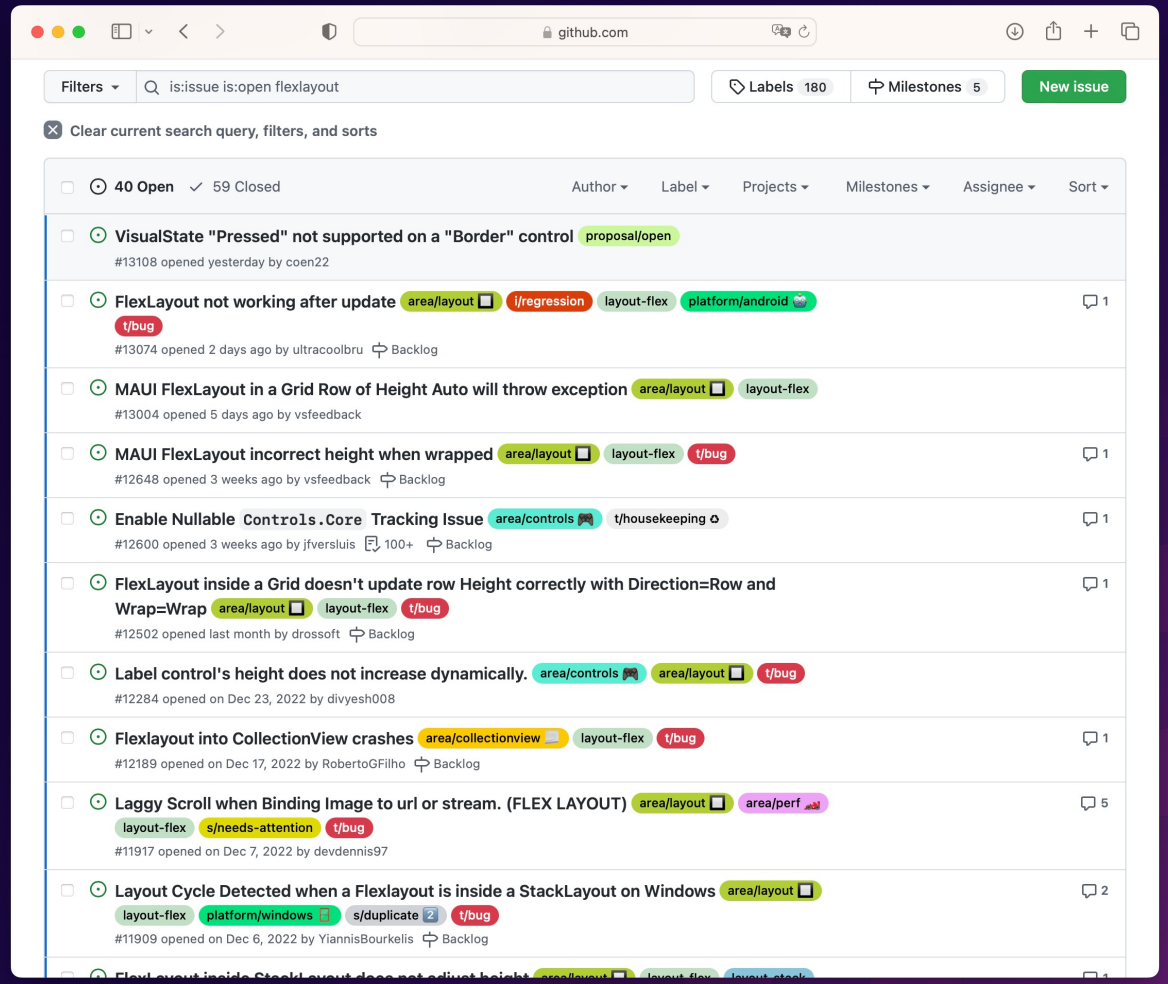
This can be converted to a `Grid` in .NET MAUI:

XAML Copy

```
<Grid RowDefinitions="Auto, *">
    <Label Text="Hello world!"/>
    <Image Grid.Row="1" Source="dotnetbot.png"/>
</StackLayout>
```

FlexLayout

- Be careful



Custom layouts

- Create an ILayoutManager and override:
  - Measure
  - ArrangeChildren

https://github.com/hartez/CustomLayoutExamples

So given this XAML:

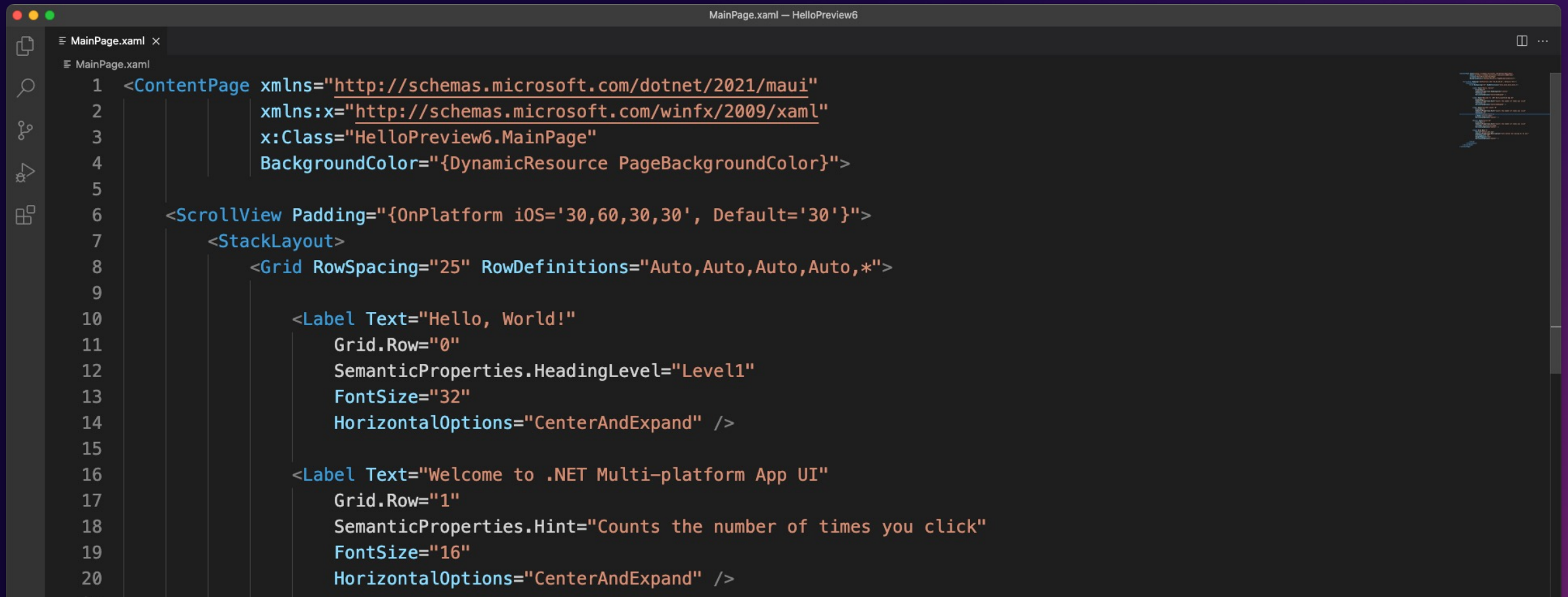
```
<local:HorizontalWrapLayout Margin="10" Spacing="5">
    <Button Text="One"></Button>
    <Button Text="Two"></Button>
    <Button Text="Three"></Button>
    <Button Text="Four"></Button>
    <Button Text="Five"></Button>
    <Button Text="Six"></Button>
    <Button Text="Seven"></Button>
    <Button Text="Eight"></Button>
    <Button Text="Nine"></Button>
    <Button Text="Ten"></Button>
</local:HorizontalWrapLayout>
```

You'll see something like this:



Xamarin.Forms to Microsoft.Maui

.NET MAUI uses the same XAML and control naming



Pre-requisites

For best success upgrading to .NET 6 we recommend:

- Upgrade any PCL projects to .NET Standard
- Upgrade Xamarin.Forms projects to version 5.0
- Upgrade packages.config to PackageReference

Recommended tools:

- Visual Studio 2022 for Windows or Mac (link)
- .NET Upgrade Assistant (link)

Custom Renderers

What's a Handler? Mapper?!



What’s a Handler? Mapper?

Migrating Custom Renderers

Scenario 1: Customizing an existing renderer

Code Blame 86 lines (70 loc) · 2.47 KB Raw

```
using CoreAnimation;
using CoreGraphics;
using SmartHotel.Clients.Core.Controls;
using SmartHotel.Clients.iOS.Renderers;
using System;
using System.ComponentModel;
using System.Linq;
using Xamarin.Forms;
using Xamarin.Forms.Platform.iOS;

[assembly: ExportRenderer(typeof(ExtendedEntry), typeof(ExtendedEntryRenderer))]
namespace SmartHotel.Clients.iOS.Renderers
{
    public class ExtendedEntryRenderer : EntryRenderer
    {
        public ExtendedEntry ExtendedEntryElement => Element as ExtendedEntry;

        protected override void OnElementChanged(ElementChangedEventArgs<Entry> e)
        {
            base.OnElementChanged(e);

            if (e.NewElement != null)
            {
                if (Control != null)
                {
                    Control.BorderStyle = UIKit.UITextBorderStyle.None;
                }

                UpdateLineColor();
                UpdateCursorColor();
            }
        }

        protected override void OnElementPropertyChanged(object sender, PropertyChangedEventArgs e)
        {
            base.OnElementPropertyChanged(sender, e);
```

Choose from among:

- AppendToMapping
- ModifyMapping
- PrependToMapping

Migrating Custom Renderers

Scenario 1: Customizing an existing renderer

```
EntryHandler.Mapper.AppendToMapping("CustomizeEntry", (handler, view) =>{
    #if IOS
    handler.PlatformView.BorderStyle = UIKit.UITextBorderStyle.None;
    handler.PlatformView.TintColor = view.TextColor.ToNativeColor();
    #endif
});
```

Migrating Custom Renderers

Scenario 2: Customizing an existing renderer with custom properties

Code Blame 17 lines (14 loc) · 457 Bytes Raw

```
using Xamarin.Forms;

namespace SmartHotel.Clients.Core.Controls
{
    public class ExtendedEntry : Entry
    {
        public static readonly BindableProperty LineColorProperty =
            BindableProperty.Create("LineColor", typeof(Color), typeof(ExtendedEntry), Color.Default

        public Color LineColor
        {
            get => (Color)GetValue(LineColorProperty);
            set => SetValue(LineColorProperty, value);
        }

    }
}
```

Create Handler and Mapper

for ExtendedEntry

AppendToMapping

Migrating Custom Renderers

Scenario 2: Reuse the Custom Renderer via Compatibility

Code Blame 17 lines (14 loc) · 457 Bytes Raw

```
using Xamarin.Forms;

namespace SmartHotel.Clients.Core.Controls
{
    public class ExtendedEntry : Entry
    {
        public static readonly BindableProperty LineColorProperty =
            BindableProperty.Create("LineColor", typeof(Color), typeof(ExtendedEntry), Color.Default

        public Color LineColor
        {
            get => (Color)GetValue(LineColorProperty);
            set => SetValue(LineColorProperty, value);
        }

    }
}
```

Create Handler and Mapper

for ExtendedEntry

AppendToMapping

Thank You!

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